

FAULT CODE 293 - Auxiliary Temperature Sensor Input 1 Circuit - Voltage Above Normal or Shorted to High Source

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ECM, all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3823993 - male Deutsch™ test lead, Part Number 3823994 - female Deutsch™ test lead, Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

CAUTION

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Codes 133 active or inactive with more than one count in the last 25 engine hours?
STEP 2.	Check the original equipment manufacturer (OEM) wiring harness.	
	STEP 2A. Inspect the OEM wiring harness and OEM interface wiring harness connector and pins.	Dirty or damaged pins?

STEPS		SPECIFICATIONS
	STEP 2B. Check for an open circuit.	Less than 10 ohms?
	STEP 2C. Check for a short circuit from pin to pin.	Greater than 100k ohms?
	STEP 2D. Check for a short circuit to engine block ground.	Greater than 100k ohms?
STEP 3.	Check the engine control module (ECM) voltage.	
	STEP 3A. Check the voltage from the ECM.	Voltage between 4.75 and 5.25 volts direct current (VDC)?
STEP 4.	OEM auxiliary temperature sensor is malfunctioning.	
	STEP 4A. A malfunctioning OEM auxiliary temperature sensor has been detected.	All other troubleshooting procedures have been completed, and the fault code is still active or inactive with more than one count within the last 25 engine hours?
STEP 5.	Check the ECM calibration and clear fault codes.	
	STEP 5A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 5B. Disable the fault code.	Fault code inactive?
	STEP .	

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for an inactive fault code. - Start the engine. Let it idle for 1 minute. - Use INSITE™ electronic service tool to read the fault code.	Fault Codes 133 active or inactive with more than 1 count in the last 25 engine hours? YES	Appropriate fault code troubleshooting symptom tree
	Fault Codes 133 active or inactive with more than 1 count in the last 25 engine hours? NO	2A

STEP 2. Check the original equipment manufacturer (OEM) wiring harness.

STEP 2A. Inspect the OEM wiring harness and the left bank OEM interface wiring harness connector pins.

Conditions : - Turn keyswitch OFF. - Disconnect the OEM wiring harness connector from the OEM auxiliary temperature sensor. - Disconnect the left bank OEM interface wiring harness connector from the ECM.		
Action	Specification/Repair	Next Step

Inspect the OEM harness and OEM temperature sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins. • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair the damaged harness, connector or pins, if possible. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	5A
	Dirty or damaged pins? NO	2B

STEP 2B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM wiring harness connector from the OEM auxiliary temperature sensor.
- Disconnect the left bank OEM interface wiring harness connector from the ECM.

Action	Specification/Repair	Next Step
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- Measure the resistance from the OEM auxiliary temperature sensor SUPPLY pin of the OEM auxiliary temperature sensor OEM wiring harness connector to the OEM auxiliary temperature sensor SUPPLY pin of the left bank OEM interface wiring harness ECM connector. - Measure the resistance from the OEM auxiliary temperature sensor RETURN pin of the OEM auxiliary temperature sensor OEM wiring harness connector to the OEM auxiliary temperature sensor RETURN pin of the left bank OEM interface wiring harness ECM connector.	Less than 10 ohms? YES	2C
	Less than 10 ohms? NO Repair : An open circuit in the OEM wiring harness has been detected. Troubleshoot the OEM wiring harness and all interconnects.	5A

STEP 2C . Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM wiring harness connector from the OEM auxiliary temperature sensor.
- Disconnect the left bank OEM interface wiring harness connector from the ECM.

Action	Specification/Repair	Next Step
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• Measure the resistance from the OEM auxiliary temperature sensor SUPPLY pin in the OEM auxiliary temperature sensor connector to all other pins in the left bank OEM interface wiring harness ECM connector. • Measure the resistance from the OEM auxiliary temperature sensor RETURN pin in the OEM auxiliary temperature sensor connector to all other pins in the left bank OEM interface wiring harness ECM connector.	More than 100k ohms? YES	2D
	More than 100k ohms? NO Repair : A short circuit in the OEM wiring harness has been detected. • Troubleshoot the OEM wiring harness and all interconnects.	5A

STEP 2D. Check for a short circuit to engine block ground.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the OEM wiring harness connector from the OEM auxiliary temperature sensor.
- Disconnect the left bank OEM interface wiring harness connector from the ECM.

Action	Specification/Repair	Next Step
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Measure the resistance from the OEM auxiliary sensor SUPPLY pin in the OEM auxiliary temperature sensor connector to engine block ground.	More than 100k ohms? YES	3A
	More than 100k ohms? NO Repair : A short circuit in the OEM wiring harness has been detected. Troubleshoot the OEM wiring harness and all interconnects.	5A

STEP 3. Check the ECM voltage.

STEP 3A. Check the voltage from the ECM.

Conditions :

- Turn keyswitch OFF.
- Disconnect the left bank OEM interface wiring harness connector from the ECM.
- Turn the keyswitch ON.

Action	Specification/Repair	Next Step

Measure the Voltage between the OEM auxiliary temperature sensor ECM SUPPLY pin at the ECM connector and the OEM auxiliary temperature sensor ECM RETURN pin at the ECM connector.	Voltage between 4.75 and 5.25 VDC? YES	4A
	Voltage between 4.75 and 5.25 VDC? NO Repair : A malfunctioning ECM has been detected. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html) in Section 19.	Repair complete

STEP 4. OEM auxiliary temperature sensor is malfunctioning.

STEP 4A. A malfunctioning OEM auxiliary temperature sensor has been detected.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM wiring harness connector from the OEM auxiliary temperature sensor.

Action	Specification/Repair	Next Step

Verify that all preceding solution verifications, for this fault code, have been performed.	All other troubleshooting procedures have been completed, and the fault code is still active or inactive with more than one count within the last 25 engine hours? YES Repair : A malfunctioning OEM auxiliary temperature sensor has been detected. Replace the OEM auxiliary temperature sensor. See equipment manufacturer service information.	5A
	All other troubleshooting procedures have been completed, and the fault code is still active or inactive with more than one count within the last 25 engine hours? NO	5A

STEP 5 . Check ECM calibration and clear fault codes.

STEP 5A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	5B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 (/qs3/pubsys2/xml/en/procedures/87/87-019-032.html) in Section 19.	5B

STEP 5B. Disable the fault code.

Conditions : - Connect all components. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete
	Fault code inactive? NO	Escalate or call for assistance

